

[INTERNAL]

Observability & Monitoring — NovaTech Platform Standards

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NovaTech adopts the three pillars of observability: metrics (Prometheus + Grafana), traces (Tempo + OpenTelemetry), and logs (CloudWatch + structured JSON). This document specifies instrumentation standards, SLO definitions, alerting runbooks, and the post-March 2026 incident improvements to detection latency.

1. Metrics Standards

All services must expose a `/metrics` endpoint in Prometheus exposition format. Required metrics: `http_request_duration_seconds` (histogram), `http_requests_total` (counter by method/status), and `process_resident_memory_bytes`. Custom business metrics (e.g., `rag_retrieval_ndcg`, `embed_cache_hit_ratio`) are defined in a shared metrics registry in the `novatech-shared-libs` package.

SLO	Target	Alert Window	Severity
API Availability	99.9%	5 min	P1
/ask P95 Latency	<500ms	10 min	P2
Embed Queue P95 Wait	<30s	5 min	P2
Qdrant Search P99	<200ms	5 min	P1
Error Rate	<0.5%	5 min	P2

2. Distributed Tracing

OpenTelemetry SDK (Python) is integrated via middleware into every FastAPI service. Traces are exported to AWS Managed Grafana Tempo. Sampling rate: 100% for error traces, 5% for success traces in production. Trace context is propagated via W3C TraceContext headers.

The March 2026 security incident was partially obscured because the Qdrant client was not instrumented with OTel. As of April 2026, all Qdrant calls emit spans with `collection_name`, `operation`, and `duration_ms` attributes.

3. Log Standards

All logs must be structured JSON with mandatory fields: timestamp (ISO 8601), level, service, correlation_id, user_id (hashed), message. PII must never appear in logs; a log scrubber middleware strips email, phone, and IP fields before emission.

- Log levels: DEBUG (dev only), INFO (business events), WARNING (degraded state), ERROR (failed operations).
- Retention: INFO and above retained 90 days in CloudWatch; ERROR retained 1 year.
- Log-based alerts: ERROR rate > 10/min over 2 minutes triggers PagerDuty P2.